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Structured teaching and classroom management – the solution for the decline of Swedish school results? Conclusions drawn from a comparative meta-synthesis of teaching and learning

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With a basis in conclusions from a comparative meta-synthesis of teaching and learning, the question of structured and teacher-led teaching in Swedish comprehensive schools is discussed and analysed. The aim is to illustrate the development of results and changes in teaching patterns in Swedish comprehensive schools in relation to new regulations in the curriculum concerning structured teaching. The result shows that in current research there is some support for structured teaching but, at the same time, a gradual and parallel development of the pupil's personal discovery and learning is emphasised. Regardless of the grade of structure, the quality of interaction and communication in teaching seems to be conclusive. One superior teaching dimension consists of *structure* and *interaction*.

Keywords: structured teaching; classroom management; comparative meta-synthesis; balanced teaching

Introduction

During the last two decades, Swedish curriculum has been characterised by a large openness in terms of the governance of teachers' teaching practice. Instead, focus has been geared towards the goals and qualities of knowledge, which the pupils are expected to achieve and develop during their time in school. The question of teachers' planning, execution and evaluation of teaching has, to a large extent, rested on the idea of a decentralised and professional responsibility to design the school's work and teaching so that the goals are achieved. Parallel with the declining results in Swedish schools, over the past decade, alleged deficiencies in teaching in Swedish comprehensive schools have now come to the forefront. The need for a clearer teacher-role and teacher-leadership in teaching has been emphasised, for example, by the Swedish Minister for Education who, in a debate, stated that teachers must now reclaim their position in the classroom. In the ongoing reformation of Swedish schools, clearer demands on teacher-led tuition have now been introduced through a new provision in the education ordinance (SFS 2011:185; The Ministry of Education, 2011a). The concept, structured teaching, which has been used in the education ordinance, has also been incorporated into the new curriculum for Swedish comprehensive schools (Lgr 11; The Ministry of Education, 2011b), where it states that

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‘School should offer pupils structured teaching under the leadership of the teacher, both individually and as part of a class’ (p. 13). The aim of this article is to, in the light of earlier research on teaching and learning:

- (i) illustrate the development of results in Swedish comprehensive school,
- (ii) discuss changing teaching patterns in Swedish comprehensive school,
- (iii) analyse how the ‘new’ education policy watchword – structured teaching – can be perceived.

In answering the aim of the article, the development of results in Swedish comprehensive school and the question of structured teaching are put in relation to aspects of teacher-leadership in the classroom, on the basis of conclusions from a review project where 16 international and 27 Swedish research reviews surrounding these questions have been analysed and synthesised.¹

The decline of school results in Swedish comprehensive schools

The discussion about the negative development of results in Swedish comprehensive schools is generally connected to pupils’ academic achievements; such as, they stem from different types of knowledge measurement and evaluation. It is against this backdrop that we see the change in curriculum mentioned above; that is, there are a number of investigations that confirm a decline in Swedish pupils’ level of knowledge within, above all, mathematics, science and reading comprehension since the mid-1990s, but also a need for possible explanations and accompanying measures (cf. Gustafsson & Yang-Hansen, 2009; Skolverket, 2009, 2010). At the same time, however, the overall picture is utterly complex with, for example, an increase in the level of grades within other subjects than those measured in international and national examinations, and, on the basis of regular surveys carried out by the Swedish National Agency for Education, increased influence and well-being for pupils in comprehensive school. However, the results that were measured from pupils’ academic achievements are often dismissed by researchers (cf. Gustafsson & Yang-Hansen, 2009) as too uncertain, and other socially and personally based results do not seem to be given equal weight in the discussion of school results. The latest PISA results in 2009 (see National Agency for Education, 2010) for the case of Sweden, also confirm an ongoing general decline within important knowledge areas. Furthermore, it appears that Sweden stands out as a country where both the average results and the equivalence have simultaneously deteriorated. The differences in results between high- and low-achieving pupils, and between boys and girls, have increased in Sweden in the last decade. The most evident deterioration is that of the lowest-achieving pupils within reading comprehension and science, while the boys’ decline in the results is equally distributed between three subject areas and stronger than the girls (Skolverket, 2010).

The causes behind the decline in results in Swedish comprehensive schools has frequently been discussed in policy, research and media contexts over the last decade. No single factor has yet been identified, even if a number of important aspects have been emphasised within and outside schools. Björklund, Fredriksson, Gustafsson, and Öckert (2010) investigate, for example, six different hypotheses and try to find reasonable explanations for the negative achievement changes in Swedish comprehensive schools between 1995 and 2008: (i) the proportion of students with

foreign background, (ii) reduction of economic resources since the beginning of 1990s, (iii) changes in resource allocation from the municipalities, (iv) increased segregation and tracking of pupils in comprehensive schools, (v) the number of well-educated teachers and (vi) changes in working forms in schools. The authors propose a combination of factors having impact on the decline of Swedish school results, primarily changes in the working forms teachers use but also changes in resource allocation and in a sense reduction of the proportion of educated teachers (cf. Björklund et al., 2010; see also Gustafsson et al., 2010 and next passage). What I initially identified in terms of new formulations in the curriculum for comprehensive school (Lgr 11; The Ministry of Education, 2011b) – the teacher's leadership within the scope of demands for a more structured teaching – is a part of what can generally be described as the importance of the educational environment for academic achievements; something that has changed in Swedish comprehensive schools over time (see next passage; cf. also Håkansson & Sundberg, 2009).

In connection with the PISA investigations, certain correlation analyses connected to the results of various countries have also been carried out. Here, positive connections emerge between pupils' academic achievements and factors related to the relationship between teacher and pupil, as well as the disciplinary climate of the school. A paradox in this context is that, during a comparison between PISA results from 2000 and 2009, both the relationship between teacher-pupil and the disciplinary climate have improved in Sweden, while, at the same time academic achievements have continued to decline. This is supported by survey responses in which the proportion of pupils who say that it is rare that the teacher must wait a long time (at the beginning of the lesson) before it gets quiet in class has risen from 57 to 71%, but also that the percentage of pupils who believe they receive extra help from the teacher when they need it has increased from 78 to 82%. However, it appears that overviews and syntheses of Swedish research, as shown in the following passage, collectively point to changes in teaching patterns that affect the teacher's control and structuring of teaching in comprehensive school; but, at the same time, these changes can be interpreted in different ways. Can the fragments and pieces of the puzzle that emerge from research on teaching lead to clear conclusions about what the school's problem is? And can they justify new curriculum formulations in which the teacher's autonomy regarding teaching is challenged; or does the map need to be both nuanced and problematised?

Changing patterns of teaching in Swedish comprehensive schools – a retrospection

Research that has studied teaching patterns and teachers' methods of organising teaching in Swedish comprehensive schools indicates significant changes during the last 20 years. Granström (2007), who has examined changes of how time is distributed between different forms of teaching (whole-class teaching, group work and individual work), indicates shifts from the mid-1990s and onwards in such a way that time for whole-class teaching and group work has decreased, while time for individual work has increased. However, other researchers (e.g. Sahlström, 2008), present studies which demonstrate that ca. 50% of the time is still being used by teachers for whole-class teaching. According to Norwegian studies (cf. Hensvold, 2006), the utilisation of time is dominated by whole-class work and individual assignments, while the time for group work constitutes a smaller proportion.

It is also evident from the Swedish research reviews that have been examined that the character of different forms of teaching has changed over time. In comprehensive school, individual work usually meant that pupils worked simultaneously with similar tasks. This form of individual work has, in recent years, however, changed character in the direction of what is called individual or personal work, where pupils personally plan and sequence the work within the scope specified by the teacher (cf. Granström, 2007). Explanations for an increase in individual orientation in school are related, in research, to social changes, amongst other things, where goals such as self-regulation, personal responsibility, independence, etc. are emphasised at the same time as research emphasises how difficult it is for pupils (certain groups in particular) to independently understand the ground rules for today's individual work. Nevertheless, there are not only negative experiences from an ever increasing share of personal work. A motivational potential in connection with pupil participation in choice of activities, planning and choice of content (see e.g. Granström, 2007; Håkansson & Sundberg, 2009) is also reported in the research.

Håkansson and Sundberg (2009), who summarise a number of Swedish studies surrounding the inner workings of schools, state that there is evidence of shifts in working forms and distribution of responsibility between teacher and pupil that have negative consequences for pupils' academic achievements. One difficulty when interpreting these research results is, however, that concepts such as individualisation are not clearly defined, for example, research emphasises favourable results on the basis of a contextual individualisation where variation in context and adaptation to pupils' varying prerequisites and needs characterise the teaching. Thus, there are certain forms of individualisation that appear to be problematic; such that, to a lesser degree, they structure pupils' learning and, in a way, leave the pupil alone to discover and understand what the teaching content is all about. However, there is international research evidence showing that lack of teacher structure and support have influence on students' school achievements, especially in relation to subject-based and easily measurable knowledge (cf. Håkansson & Sundberg, 2012b). Research from an English context also shows that transition to less structured working forms to a higher extent affects pupils with less developed study qualifications. This group seems to benefit from explicit and more structured literacy teaching in terms of improving performance levels of literacy skills (cf. Machin & McNally, 2008; see also Björklund et al., 2010).

On the basis of this analysis of research that over time has examined changing patterns in teaching in Swedish comprehensive schools, formulations concerning structured teaching that appear in comprehensive school curriculum and the education ordinance, could be nuanced and problematised. According to research, there is still much teacher-led teaching in Swedish classrooms, even if the proportion of individual work without adequate teacher support is thought to have increased. The question is how structured the teaching is; how it is led by the teacher; and how good the quality of communication is – factors that have proved important for the development of learning in pupils (cf. Håkansson & Sundberg, 2012a). Another question that is strangely absent in the Swedish discussion on quality in teaching and school results is how pupils' group, and cooperative, learning is used in teaching; something that, in international research, is an extensive research field (cf. e.g. Slavin, 2011). For the continued understanding of research positions and results within the field of teaching and learning, the named project is presented in the following section on methodology.

A comparative meta-synthesis on teaching and learning

Reviews of research within different scientific fields has long been an established form of compilations of existing research in the form of research, knowledge and literature reviews; for example, meta-analyses, narrative reviews and meta-ethnographies. In recent decades, the term systematic research review has been used within the field of education as a characteristic for reviews that use a developed and transparent methodology. In a systematic research review, work is based on defined questions, search strategies and search criteria, amongst other things; but also many predefined steps for data extraction and comparative analyses (cf. Oakley, Gough, Oliver, & Thomas, 2005; Sundberg, 2009). In a comparative meta-synthesis (Håkansson & Sundberg, 2012a), both international and Swedish teaching and learning research are compared with the help of data from 16 international and 27 Swedish research reviews. The comparative meta-synthesis is an example of yet another variant within the systematic research reviews; in this case, many types of reviews are used as empirical material within the field of teaching and learning. Such a review does not work with summarising effect sizes as, for example, meta-analyses do, rather it has the ambition to create logical inference chains from different sources in order to gradually strengthen the comparative analyses of different research reviews (cf. e.g. McCormick, Rodney, & Varcoe, 2003). Both quantitative and qualitative research can be included, depending on how it responds to the review questions. The aim of Håkansson and Sundberg's (2012a) comparative and qualitative meta-synthesis has been to: (i) identify dominant research positions and trends within the field of teaching and learning, (ii) compare results as they appear in analyses of international and Swedish reviews as well as and (iii) reconstruct and draw conclusions on the basis of these analyses.

Methodology in the comparative meta-synthesis

One feature of the comparative and qualitative meta-synthesis is the concept, translation, which should not be taken literally but rather in the sense of translation of the text's sentence. In the synthetisation of reviews surrounding the field of teaching and learning, Noblit and Hare (1988), who describe three possible forms of translation, have followed:

- (i) reviews are directly comparable and it is possible to carry out mutual sentence translations,
- (ii) reviews are in relative opposition to one another and repelling, and
- (iii) reviews can be combined to form a line of argument; a series of inferences.

The reviews contained in the empirical analysis have been considered as cases, and synthesis has been carried out in the attempt to find patterns of variation between different cases. Through gradually analysing the different reviews, the most important factors have been identified and a generalisation pattern of dividing lines and concurrences in the research field have emerged. In this way, a nuanced map of key factors and components, which interact in research positions and results in the field of teaching and learning, have been reconstructed (cf. Håkansson & Sundberg, 2012a).

The concrete conducting of analyses has followed seven methodological steps where the first three dealt with identifying interests for the review (aim, focus and demarcations), formulating inclusion and exclusion criteria based on identified interest as well as formulating relevance criteria, for example, that the reviews should take teacher, pupil and contextual aspects into account (cf. Danish Clearinghouse for Educational Research [DCU], 2006; Noblit & Hare, 1988; Thorne, Jensen, Kearney, Noblit, & Margarete, 2004².) The choice of international research reviews was made on the basis of the following criteria:

- (i) Key words within the research field of didactics/teaching and learning.
- (ii) Reviews of research.
- (iii) Publication years 1990–2010.
- (iv) International in orientation and scope.
- (v) Claim to (re-)present the current front in international research in the field.
- (vi) Written in English and corresponding to the formal education system.
- (vii) Focus on general and generalisable empirical research results.
- (viii) Represented in the major international educational databases, international educational publishers and/or the international organisations within the educational field.

The selection of the Swedish reviews was also based on the selection criteria above (excluding iv, vii, viii). Systematic literature retrieval was carried out in relevant Swedish databases and via a personal database that was established through earlier review work (Håkansson & Sundberg, 2009), but the choice could not be carried out as strictly in the Swedish field as in the international one. The reason for this is that the Swedish release of systematic reviews, thus far, appears to be relatively limited and in certain ways methodologically underdeveloped; this is why a handful of publications that can be regarded as anthologies and review articles instead of systematic reviews have also been included in the choice. This means that the selection of certain references in the Swedish choice has partly been based on the above selection criteria; however, it has also been built on qualitative assessments in relation to the research interest for the comparative meta-review.³

Included in the following two steps (4 and 5) were reading, scanning and protocol writing as well as translation, where descriptive reports, in accordance with established schedules, laid the foundations for extraction and composition of analytical themes in an analytical framework. In the matter of research on student learning, we identified for example several possible categories for presenting the empirical results, but we had to test, consider and reconsider by moving back and forth between descriptive reports, stepwise interpretations and tentative analytical themes. The categories for covering both Swedish and international research results in the case of student learning were: (i) the multicontextual childhood, (ii) the importance of pre-school, (iii) research on student learning – an overview, (iv) learning theories, (v) the importance of classroom communication, (vi) subject content knowledge and generic skills, (vii) the Swedish research on subject didactics and (viii) motivation, goal-orientation and social bonds (Håkansson & Sundberg, 2012b). The sixth step in working with the qualitative meta-synthesis was made up of the synthetisation of the translations; a form of comparative analyses of the interpretations that were made in step 5 in the direction of an integrative reconstruction. A methodological reflection is that there is still a need to develop the explicit reports of the chains of arguments

(from the empirical material in the research reviews) leading up to partial analysis, whole analysis and conclusions. Even if it is not possible to show in every detail how the running interpretation and analysis is completed, there is room for development work with improvement of the documenting of the comparisons (Håkansson & Sundberg, 2012a). The seventh step – communication of synthetisation – which this article is an example of, deals with attempting to translate the qualitative meta-review to the discourses to which the review is thought to contribute; for example, journals and other types of publications that can act as bridge builders and collective reconstructions in order to understand and explain the development of the research field in dialogue with researcher and practitioner (Håkansson & Sundberg, 2012a).⁴

General results from the comparative meta-synthesis

The comparisons and analysis of the 43 research reviews led to a broad picture – a map – from the development of the research field teaching and learning. Of certain interest here however is the mapping of empirical results from the selected research. There are many ways of describing this huge amount of research for example in terms of what influences pupil achievement inside and outside schools (see for example Hattie, 2009). In our case, the starting point for the analysis is the so-called didactic triangle (teacher–pupil–content/teaching method) and the context of classrooms (cf. Håkansson & Sundberg, 2012b). The general map from the analysis contains results from research on teaching in terms of: (i) general principles for excellent teaching, (ii) classical questions (class-size, differentiation, gender and multicultural teaching), (iii) students learning, (iv) teacher practice and (v) certain teaching practices: assessment, special education and ICT.

One of the most obvious result from the meta-review with regard to factors inside schools, is the importance of the teacher from many different aspects (point iv above). In our categorisation of research on teachers and teaching, we identify at least eight domains of importance for pupils learning and the question of excellent teaching:

- (1) Teachers' knowledge base.
- (2) Teachers' subject matter knowledge.
- (3) Teachers' values and attitudes.
- (4) Teachers' expectations and labelling.
- (5) Teachers' relationships and feelings.
- (6) Teachers using effective teaching strategies.
- (7) Classroom management.
- (8) Evaluation and certification of teachers.

Since my main focus in this text is the question of how structured teaching can be put in relation to aspects of teacher-leadership in the classroom, I have used findings from the meta-synthesis above all related to points 6 and 7 mentioned above.

Structured teaching and teacher-leadership in teaching

The concept, structured teaching, which has appeared in the curriculum and regulations for Swedish comprehensive schools, can be a part of a number of other aspects, related to research within the field of teaching and learning that has been studied and analysed. In the curriculum, the concept, structured teaching, is followed

by the formulation ‘... under the leadership of the teacher’, which indicates a steering in the direction of a clearer leader-role for the teacher during tuition. According to Swedish regulations, however, it is unclear as to what distinguishes teaching from structured teaching, as, in the Education Act (Sverige, 2010, 1 chap. 3§), it already states that teaching implies ‘... such goal-oriented processes that, *under the leadership of the teacher or preschool teacher*, aim to develop and teach through the acquisition and development of knowledge and values’ (the author’s italics). The purpose behind the introduction of the ‘new’ motto, structured teaching, seems, after all, to be some form of emphasis and reinforcement of the teacher-role – a teacher-role that is said to have been toned down concurrently with the utilisation of more and more individual working forms for pupils.

Even if – as presented earlier – whole-class teaching in Swedish comprehensive schools is, according to research studies, relatively common, the Swedish Minister for Education uses the lack of it as an argument for the introduction of regulations on structured teaching. What is more, the minister connects the term *pulpit teaching* (recitation) with structured teaching:

The long hated ‘pulpit teaching’ must once again become common in Swedish classrooms. Teacher-led teaching does not only involve the teacher going through the material, explaining, instructing and repeating, but also the teacher having an active dialogue with an entire class of pupils where questions and problems are discussed. (Dagens Nyheter, 2011)

So how do this reasoning relate to what research reviews within the field of teaching have arrived at in recent years? What actual support is there in the research for a greater emphasis on teacher-leadership in education? In a recent more extensive research handbook within the field of teachers and teaching (Saha & Dworkin, 2009), the term, classroom management, is extensively discussed. Barbara Landau, who summarises the state of research, distinguishes between a behaviouristic and a constructivist or democratic approach to classroom management, where the first named is strongly teacher-led, without pupil influence and passivity and obedience as typical behaviour, while the latter builds on the teacher and pupils developing rules together, which are based on common values and principles. Today’s curricula in different countries lean more and more towards a constructivist approach, but according to Landau, this has sometimes been misunderstood; for example, that pupils would completely dominate decisions and applications of rules. The conclusion is that it is fundamental to have some form of management in the classroom that is built on reciprocity and verbal encouragement from the teacher in order to support suitable decisions and self-regulation, rather than an emphasis on punishment and rewards for different types of behaviour. Landau argues that leadership in the classroom must focus on things that best support pupils’ development towards becoming respectful adults (cf. Landau, 2009).

Effects of teaching methods – the example of classroom management

A branch of the research on teaching that has received a lot of attention in recent years is effect research on teachers and their teaching. In many international reviews and meta-analyses, conclusions are mainly drawn from quantitative research that relates to a number of different influential factors for pupils’ achievement. In Nordenbo, Sögaard Larsen, Tiftikci, Wendt, and Östergard (2008), a narrative

conclusion is drawn from 27 of the studies that observe the question regarding the importance of classroom management for pupils' study representation. The principle message, when it comes to teacher-leadership in the classroom, is that in order to facilitate pupils' study representation, the teacher needs to concentrate on:

Careful planning – if more time is dedicated to teaching (WHAT) compared to the time that is dedicated to administrative routines (HOW), this will favour the pupils' learning.

Clear teaching goals – if the teacher formulates clear short-term and long-term goals that the pupil understands and accepts, this will be advantageous to pupils' learning.

Pupil-support leadership – if pupils are involved in the structuring and choice of activities in the class, and if they are given the opportunity to take responsibility, this will promote their learning.

Activation and motivation of pupils – there are connections between classroom management and pupils' self-regulated behaviour in relation to pupils' inner motivation. It is reported in research, for example, that pupils with cognitive-oriented teachers develop a higher level of self-regulation than pupils who are taught by teachers who focus their classroom management on pupils' behaviour.

Organisation of activities and learning – several studies point out that planned teaching, which secures the connection between previously discussed content and progression in learning, favours pupils' learning. This occurs, for example, when the teacher helps pupils to focus on central parts of the curriculum, follow up earlier lessons, provide fast and correct feedback and emphasise important principles in the knowledge content.

Visible leadership – according to studies carried out by Nordenbo and his colleagues, the levels of pupils' learning are thought to increase when the teacher appears as a clear leader. A visible leadership does not contradict other valuable dimensions that have been mentioned above, such as, activation, support and influence for pupils. (cf. Nordenbo et al., 2008, pp. 51–52)

In connection with studies that deal with classroom management, research that concerns *teachers' leadership of pupil behaviour* is also dealt with; and on a more extensive level, studies that highlight teachers' work with the psychological climate in the classroom – *the classroom climate*. According to Danish researchers, the label, classroom climate, is a comprehensive term that, in reality, is equal to such factors as those named above in connection to classroom leadership including good social relationships with pupils. Classroom management, or *rule management competence* as the authors like to call it, is, in combination with teachers' *relational competence* and *didactic competence*, the most important aspect in contributing to pupils' achievement. Naturally, within the latter aspects are a number of important dimensions worth paying attention to, but that do not fit within the scope of this text (cf. Nordenbo et al., 2008).

Hattie (2009), who in a vastly extensive research review⁵ has calculated the effects of different influential factors on pupils' academic achievements, and has also highlighted a number of factors that can be related to classroom management in accordance with the above-mentioned Danish review; for example, studies leadership in the classroom, clarity of the teacher surrounding goals and knowledge in teaching and trustful relationships between teacher and pupil. According to Hattie's method of ranking these factors on the basis of their effect sizes, they are important in relation to everything else that influences pupils' academic achievements. Hattie also mentions the classroom climate as the umbrella term in the context, where,

aside from the mentioned factors, the teacher's ability to reduce disturbing behaviour in the classroom is also emphasised as important for pupils' learning.

Structured teaching – interpretation and analysis of the curriculum in relation to research on teaching and learning

Have these aspects of effective teaching that have been selected here something to do with the terms structured teaching or pulpit teaching? Many believe they have. Nevertheless, the question is more complex than this. Much depends, quite simple, on how we define pulpit teaching or structured teaching. According to the National Encyclopaedia (gathered 11 October 2011 from www.ne.se), pulpit teaching is defined as:

... also called classroom teaching, the traditional teaching form in school that is headed by the teacher through lectures, demonstrations and examinations with the class collectively. Contrasts are individualisation and group work, where the teacher communicates with pupils individually or in smaller groups. Contrasts are also working school and activity education, which means the students themselves are involved in the choice of educational material and method.

An initial interpretation of the research field, on the basis of the classroom management aspect, points in the direction of some form of balance between the poles described in the National Encyclopaedia explanation. On the basis of research on teaching qualities, it is important that not just the teacher is in control; not just the pupil either for that matter, but rather a form of well-planned interplay between the teacher and pupil, framed by a good classroom climate that is characterised by various teaching repertoires, which lead to learning.

It does not seem reasonable that, in the encyclopaedia, structured teaching should be tantamount with pulpit teaching, nor from the minister's argument; even if the two concepts are used there side by side. In English language research, the concept, 'direct instruction', is sometimes used to illustrate teaching sequences that contain transparent and clear goals, exemplifications of critical aspects in the content, supervision, control of understanding, summaries and repetitions (cf. Hattie, 2009, pp. 204–207). Another interpretation is that the concepts, 'direct instruction' and structured teaching have something to do with each other, that is, the teacher's role is clear but it requires the pupil's participation and includes many possible organisational forms of the class or group such as whole-class teaching, group teaching, working in pairs or individual work. Pulpit teaching, in the more traditional sense, is thought to correspond better to other English expressions such as 'teacher-centred teaching' 'recitation' or 'didactic teacher-led talking from the front'. In relation to the relationship between 'recitation' and more varied models of teaching, Joyce and Calhoun (2009) state that: 'When studies of models of teaching compare students taught with various models with students taught with the recitation, the recitation invariably loses' (Joyce & Calhoun, 2009, p. 651).

Finally, the interpretation is that structured teaching does not rule out different strategies for the treatment of knowledge content, for example, subject teaching, projects or themed teaching. Structured teaching can be included in these, both entirely and partly. Explained in a Swedish research project concerning project and theme organised teaching in secondary and upper secondary school are the difficulties of presenting different types of classroom activities, such as homogeneous or possibly distinguishing between project and pulpit-based teaching (Säljö, Jakobsson, Lilja, Mäkitalo, & Åberg, 2011). Preferably, a project-oriented teaching can be said to

accommodate varied educational activities and assessment methods where the teacher's control and management also varies according to project phases and the emerging needs of individual students and groups of students (Säljö et al., 2011).

Balance and variation – the key words in teaching

In analyses of both Swedish and international research on teaching and learning, it is obvious that neither simple solutions nor individual factors emerge as answers to the questions regarding ways in which teaching can best be planned, carried out and evaluated. Thus, it seems unlikely that the curriculum provision of structured education as individual action would have decisive effects on the performance of Swedish comprehensive schools.⁶ The teacher's leadership in teaching is one of many qualities that interact with a number of others in a focused context. There are further limitations and possibilities, if, for example, we take a look at what is happening outside of school, in the home, during leisure time, in the political sphere and in the social sphere. If, at this point, we concentrate on the school context and the concept of structured teaching, we can see that the words, balance and variation, are considered important when we discuss different ways of dealing with teaching. A too far-reaching individualisation (e.g. personal work without teacher support) has no research support, but, on the other hand, neither does a one-sided pulpit teaching. In a review, the OECD organisation CERI (Centre for Educational Research and Innovation) has, amongst other things, clarified the state of research in regard to the relationships between the social constructivist perspectives on teaching and more teacher-led teaching principles (Dumont, Istance, & Benavides, 2010). The results that emerge there speak for a good balance between discovery and personal investigation, on the one hand, and systematic teaching and guidance, with consideration taken for individual differences in abilities, needs and motivation amongst the pupils, on the other. The individual constructivism that appears to have emerged in the Swedish classroom, in the light of the previous curriculum forms in the beginning of the 1990s, is challenged by the gathered teaching and learning research (Håkansson & Sundberg, 2012a). Consequently, it is not about emphasising an entirely teacher-led teaching but, rather, balancing these approaches over time. Pupil balance is led, in this case, by how personal competence and the ability for self-regulation develop. When such individual abilities increase, the expressed and structured teaching can correspondently decrease in size (Dumont et al., 2010, pp. 53–54).

In a case study on a comprehensive school (years 6–9), the above-mentioned relationships between structure and personal discovery in teaching have been identified (cf. Adolfsson, Håkansson, & Sundberg, 2011). The teachers in the school have undergone in-job training surrounding entrepreneurial learning and work with developing this with the objective of increasing pupils' motivation for schoolwork. One question that keeps the teachers occupied in this work is, for example, how the balance between the teachers' and the pupils' control of teaching can best be maintained, but also how the balance between more interdisciplinary skills in pupils (creativity, problem-solving abilities, etc.) and subject-specific knowledge should look. Here, the question of classroom leadership includes aspects and strains surrounding both the role of the teacher and the organisation of knowledge content. What is more, the results of the study point to another aspect connected to the others; namely, assessment where teachers need to relate to both a formative and a

summative organisation of the assessment, that is, the organisation of teaching has consequences for the assessment forms (Adolfsson et al., 2011).

Summary and conclusion

It can be summarised that the regulations regarding structured teaching, which now apply in Swedish comprehensive schools, are not entirely easy to interpret. There is still no established definition of the concept, even if certain clues are given in the general recommendations by the Swedish National Agency for Education with comments on the planning and carrying out of teaching (Skolverket, 2011). I have not come across especially well-established explanations for what, in practical contexts, can be regarded as structured teaching. Thus, a 'new' education policy (or pedagogical) watchword needs to be nuanced and defined in order to contribute to the development of Swedish comprehensive schools. What is more, proud proclamations and curricula need to be problematised and studied empirically. From the research that has been referred to here, the conclusion can be drawn that some support for structured teaching by the teacher has been identified; however, at the same time, a gradual and parallel development of pupils' personal discovery and learning is emphasised. In this presentation, classroom leadership and the work of the teacher have been emphasised, which means that the pupil's perspective, but also other more comprehensive perspectives, have been toned down due to space. It is clear from the research that we cannot ignore that all teaching requires relations and interplay between teacher and pupil, but also with knowledge and values in a societal and local context. If, for example, the pupil does not see the point of the teaching, does not understand the goals, cannot be bothered to learn and so on, it is likely that the teacher's structure and leadership of teaching will not make much of a difference. This is the case for all teaching, to some extent context-bound; which means that general principles that are highlighted in research always need to be designed to the situation in relation to pupils' needs and understanding.

In our analysis of international and Swedish reviews surrounding teaching and learning, a number of important aspects, regarding the relationship between teacher and pupil, can be identified (Håkansson & Sundberg, 2012a). The quality of interplay and communication between teacher and pupil is considered to be crucial. Another conclusion to draw is that the teacher's leadership in teaching is one of the many important and interactive sub-aspects in the context. When analysing the research support for a concept like structured teaching, it is possible to identify a superior dimension for teaching quality in terms of *structure* and reciprocal *interaction*, which respond to both a contextual and social as well as organisational side of teaching (cf. Alexander, 2010; Hattie, 2009; Nordenbo et al., 2008). Context-wise, in terms of well-organised and hierarchically ordered knowledge structures; socially and organisationally, in terms of the main focus of this article: (i) coherent and appropriate forms of instruction in different groups that create an educational structure, (ii) the educational use of time and resources directed towards knowledge content rather than on administrative needs, and (iii) the construction and management of a communicative, supportive and constructive interaction between teacher and pupil and between pupils (cf. Håkansson & Sundberg, 2012a, 2012b).

Notes

1. CARL – Comparative Analysis of Research on Teaching and Learning (Håkansson & Sundberg, 2012a).
2. DCU stands for Danish Clearinghouse for Educational Research; a unit at Denmark's educational university, which works with developing systematic research reviews within the educational field on the basis of the latest available research. The so-called 'koncept-notat' the research group developed for working with systematic reviews has been the inspiration for our '7 steps model' together with Thorne et al., (2004) and Noblit and Hare (1988).
3. A list of the research reviews included is presented in Appendix 1.
4. The critical question on the relevance of systematic research reviews within the educational field, often connected to method and evidence discussions, will not be held explicitly in this text (see further, e.g. Bohlin, 2010; Davies, 2000; Evaldsson & Nilholm, 2010; Krejsler, 2011).
5. Visible Learning is a meta-metasyntesis of 800 meta-analyses based on 50,000 studies with ca. 80 million pupils.
6. What is more, the concept, structured teaching should be regarded as a tautology, as the word, teaching, already accomodates teacher guidance of goal-oriented processes (cf. Sverige, 2010, 1 chap. 3§).

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Appendix 1

Table A1. List of the international research reviews included in the meta-synthesis (in chronological order).

Author, title, year	Number of pages
Shulman, Lee S. (Ed.) (1990). <i>Research in teaching and learning: A project of the American Educational Research Association. Vol. 1, Paradigms and programs</i> . New York: Macmillan	96
Erickson, Frederick, Linn, Robert. L (1990). <i>Research in teaching and learning: A project of the American Educational Research Association. Vol. 2, Quantitative methods, qualitative methods</i> . New York: Macmillan	200
Biddle, Bruce J., Good, Thomas L. & Goodson, Ivor F. (Eds.) (1997). <i>International handbook of teachers and teaching</i> . Dordrecht: Kluwer	1474
Richardson, Virginia (Ed.) (2001). <i>Handbook of research on teaching</i> (4th ed.). Washington, DC: American Educational Research Association	1278
Gayle, Barbara Mae. (Ed.) (2006). <i>Classroom communication and instructional processes: Advances through meta-analysis</i> . Mahwah, NJ: Lawrence Erlbaum Associates	451
Alexander, Patricia A. & Winne, Philip H. (Eds.) (2006). <i>Handbook of educational psychology</i> (2nd ed.). Mahwah, NJ: Erlbaum	1055
Higgins, S. Baumfield, V. Hall, E. (2007). Learning skills and the development of learning capabilities. Report. In: <i>Research evidence in education library</i> . London: EPPI-centre, Social Science Research Unit, Institute of Education, University of London. http://eppi.ioe.ac.uk/cms/Default.aspx?tabid=1851	28

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Table A1. (Continued).

Author, title, year	Number of pages
Scheerens, Jaap et al. (2007). Review and meta-analyses of school and teaching effectiveness. www.iqb.huberlin.de/lehre/dateien/rapportScherens.pdf	221
Nordenbo, Sven-Erik et al. (2008). Teacher competences and pupil learning in pre-school and school – A systematic review carried out for the Ministry of Education and Research Oslo. In: <i>Evidence base</i> . Copenhagen: Danish Clearinghouse for Educational Research, School of Education, University of Aarhus.	101
Hattie, John A. C. (2008). <i>Visible learning: A synthesis of over 800 meta-analyses relating to achievement</i> . London, New York: Routledge	378
Florian, Lani (Ed.) (2009). <i>The Sage handbook of special education</i> London: SAGE	580
Voogt, Joke; Knezek, Gerald (Eds.) (2009). <i>International handbook of information technology in primary and secondary education</i> Series: Springer International Handbooks of Education, Vol. 20	538
Saha, Lawrence J. & Dworkin, A. Gary (Eds.) (2009). <i>International handbook of research on teachers and teaching</i> . P. 1. New York: Springer	642
Saha, Lawrence J. & Dworkin, A. Gary (Eds.) (2009). <i>International handbook of research on teachers and teaching</i> . P. 2. New York: Springer	577
Alexander, Robin J. (Ed.) (2010). <i>Children, their world, their education: The final report and recommendations of the Cambridge Primary Review</i> . New York: Routledge	577
Andrade, Heidi L. & Cizek, Gregory J. (Ed.) (2010). <i>Handbook of formative assessment</i> . New York: Routledge	377
Total: 16 international research reviews included	8035 pages

Table A2. List of the Swedish research reviews included in the meta-synthesis (in alphabetical order).

Author, title, year	Number of pages
Arfwedsson, Gerhard (1994). <i>Nyare forskning om lärare – presentation och kritisk analys av huvudlinjer i de senaste decenniernas engelskspråkiga lärarforskning</i> . Stockholm: HLS Förlag.	159
Bengtsson, Jan (1997). Didaktiska dimensioner. Möjligheter och gränser för en integrerad didaktik. In: <i>Pedagogisk Forskning i Sverige</i> , årg. 2 nr 4 s. 241–261 issn. 1401–6788	20
Berg, Gunnar & Scherp, Hans-Åke (Eds.) (2003). <i>Skolutvecklingens många ansikten</i> . Stockholm: Myndigheten för skolutveckling	299
Björkqvist, Ole (Ed.) (2005). <i>Lära ut och in: om innehållet i pedagogisk verksamhet</i> . Stockholm: Vetenskapsrådet Tillgänglig på Internet: http://www.vr.se/publikationer/sida.jsp?resourceId=-132	61
Bunar, Nihad (2010). <i>Nyanlända och lärande: en forskningsöversikt om nyanlända elever i den svenska skolan</i> . Stockholm: Vetenskapsrådet Tillgänglig på Internet: http://www.cm.se/webbshop_vr/pdf/2010_06.pdf	127
Carlgrén, Ingrid, Forsberg, Eva & Lindberg, Viveca (2009). <i>Perspektiv på den svenska skolans kunskapsdiskussion</i> . Stockholm: Stockholms universitets förlag	146
Englund, Tomas (2004). Nya tendenser inom pedagogikdisciplinen under de tre senaste decennierna. I <i>Pedagogisk Forskning i Sverige</i> 2004 årg 9 nr 1 s 37–49 issn 1401–6788 .	13
Forsberg, Eva (2010). <i>Svensk forskning om bedömning: en kartläggning</i> . Stockholm: Vetenskapsrådet	141
Granström, Kjell & Einarsson, Charlotta (1995). <i>Forskning om liv och arbete i svenska klassrum: en översikt</i> . Stockholm: Statens skolverk Tillgänglig på Internet: http://www.skolverket.se/publikationer?id=48	85
Granström, Kjell (2003). Arbetsformer och dynamik i klassrummet. In Staffan Selander (Ed.), <i>Kobran, nallen och majjen: tradition och förnyelse i svensk skola och skolforskning</i> . Stockholm: Myndigheten för skolutveckling	22
Granström, Kjell (Ed.) (2007). <i>Forskning om lärares arbete i klassrummet</i> . Stockholm: Myndigheten för skolutveckling	270
Helldén, Gustav, Lindahl, Britt & Redfors, Andreas (2005). <i>Lärande och undervisning i naturvetenskap: en forskningsöversikt</i> . Stockholm: Vetenskapsrådet	127

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Table A2. (Continued).

Author, title, year	Number of pages
Hensvold, Inger (2006). <i>Elevaktiva arbetsmodeller och lärande i grundskolan: en kunskapsöversikt</i> . Stockholm: Myndigheten för skolutveckling	154
Härnqvist, Kjell, Achtenhagen, Frank, Rosengren, Karl Erik & Öhngren, Bo (Eds.) (1997). <i>An evaluation of Swedish research in education</i> . Stockholm: Swedish Council for Research in the Humanities and Social Sciences (Humanistisk-samhällsvetenskapliga forskningsrådet), (HSFR)	287
<i>International evaluation of didactics: Swedish, mathematics and natural science</i> . (2006). Stockholm: Vetenskapsrådet	62
Lundgren, Ulf P., Säljö, Roger & Liberg, Caroline (Eds.) (2010). <i>Lärande, skola, bildning: [grundbok för lärare]</i> . 1. utg. Stockholm: Natur & kultur	630
Melander, Helen & Sahlström, Fritjof (2010). <i>Lärande i interaktion</i> . 1. uppl. Stockholm: Liber	264
Nilholm, Claes & Björck-Åkesson, Eva (Eds.) (2007). <i>Reflektioner kring specialpedagogik: sex professorer om forskningsområdet och forskningsfronterna</i> . Stockholm: Vetenskapsrådet Tillgänglig på Internet: http://www.cm.se/webbshop_vr/pdf/VR2007_05.pdf	114
Persson, Sven (2008). <i>Forskning om villkor för yngre barns lärande i förskola, förskoleklass och fritidshem</i> . Stockholm: Vetenskapsrådet Tillgänglig på Internet: http://www.cm.se/webbshop_vr/pdf/VR2008_11.pdf	119
Sahlström, Fritjof (2008). <i>Från lärare till elever, från undervisning till lärande: Några utvecklingslinjer i klassrumsforskningen</i> . Stockholm, Vetenskapsrådet.	85
Skolverket (2007). <i>Vad händer med läsningen: En kunskapsöversikt om läsundervisningen i Sverige 1995–2007</i> , Stockholm: Skolverket, Fritzes [distributör].	151
Skolverket (2008). <i>Vad händer i NO-undervisningen? En kunskapsöversikt om undervisningen i naturorienterade ämnen i svensk grundskola 1992–2008</i> . Stockholm: Skolverket	126
Skolverket (2009). <i>Vad påverkar resultaten i grundskolan. En kunskapsöversikt om betydelsen av olika faktorer</i> . Stockholm: Fritzes.	264
Skolverket (2010). <i>Perspektiv på barndom och barns lärande. En kunskapsöversikt om lärande i förskolan och grundskolans tidigare år</i> . Stockholm: Skolverket. Tillgänglig på Internet: http://www.skolverket.se/publikationer?id=2393	239

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Table A2. (Continued).

Author, title, year	Number of pages
Schüllerqvist, Bengt, Ullström, Margaretha & Ullström, Sten-Olof (Eds.). (2009). <i>Ämnesdidaktiska brobyggen – didaktiska perspektiv inom lärande och forskning</i> . Karlstad: Karlstad University Press.	251
Säljö, Roger (2003). Föreställningar om lärande och tidsandan. In Staffan Selander (Ed.), <i>Kobran, nallen och majjen: tradition och förnyelse i svensk skola och skolforskning</i> . Stockholm: Myndigheten för skolutveckling	20
Wallby, Karin, Carlsson, Synnöve & Nyström, Peter (2001). <i>Elevgrupperingar: en kunskapsöversikt med fokus på matematikundervisning</i> . Stockholm: Skolverket Tillgänglig på Internet: http://www.skolverket.se/publikationer?id=864	178
Total: 27 Swedish research reviews included	4414 pages